AI INSTITUTE #AIISC
UNIVERSITY OF SOUTH CAROLINA

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CSCE 190: Computing in the Modern World

Class 7 – Testing (Solving with Computing)

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

OCT 2026

Carolinian Creed: "I will practice personal and academic integrity."

Credits: **Copyrights of all material reused acknowledged**

Outline

- Introduction Section
  - Recap from Last Class
- Main Section
  - Problem Solving Process
  - **Validating the Problem**
    - Creating Requirements and Specifications
    - **Testing – Concepts and Activities**
  - Selecting Problems that Matter
- Conclusion Section
  - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 7

We talked about

- Problem Solving Process
- **Validating the Problem**
  - **Creating Requirements and Specifications**
- Selecting Problems that Matter

Assignment 7: Same for All in a Team

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Write a problem statement
  - The group will have one problem statement
  - Add your statement to your GitHub in a sub-folder called ‘assignment-6’
    - Do mention all the students in your group

- Write for **client**
  - **Requirements**: what will meet the need?
- Write for **developers**
  - **Specification**: what will the system-to-be built do
  - **Test plan**: how will we show that the built system works to meet the specification
- Add requirements, specification, and test plan to your Github in a sub-folder called ‘assignment-7’
  - Do mention all the students in your group

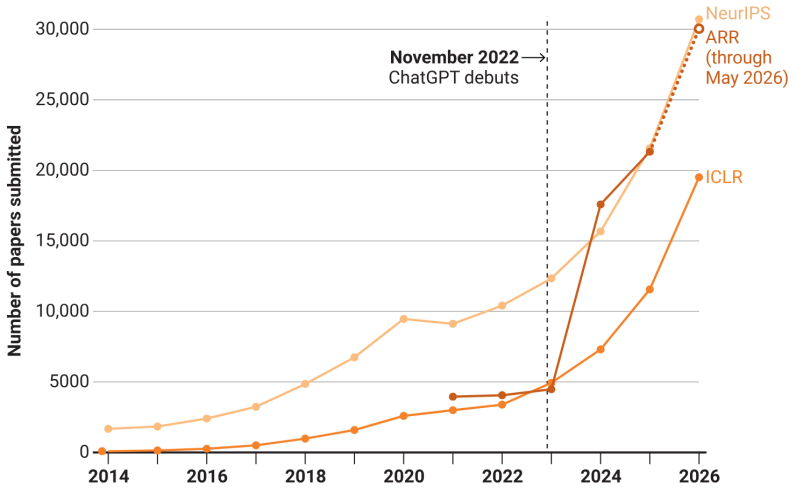
See samples at: https://github.com/biplav-s/course-cmw-f26/tree/main/software-dev

Computing in the News

AI slop tsunami, Oct 2026, https://www.science.org/content/article/ai-triggers-tsunami-submissions-computer-science-conferences-impose-caps

Arxiv received 40,363 submissions in September 2026, nearly double the tally of 20,569 in September 2024

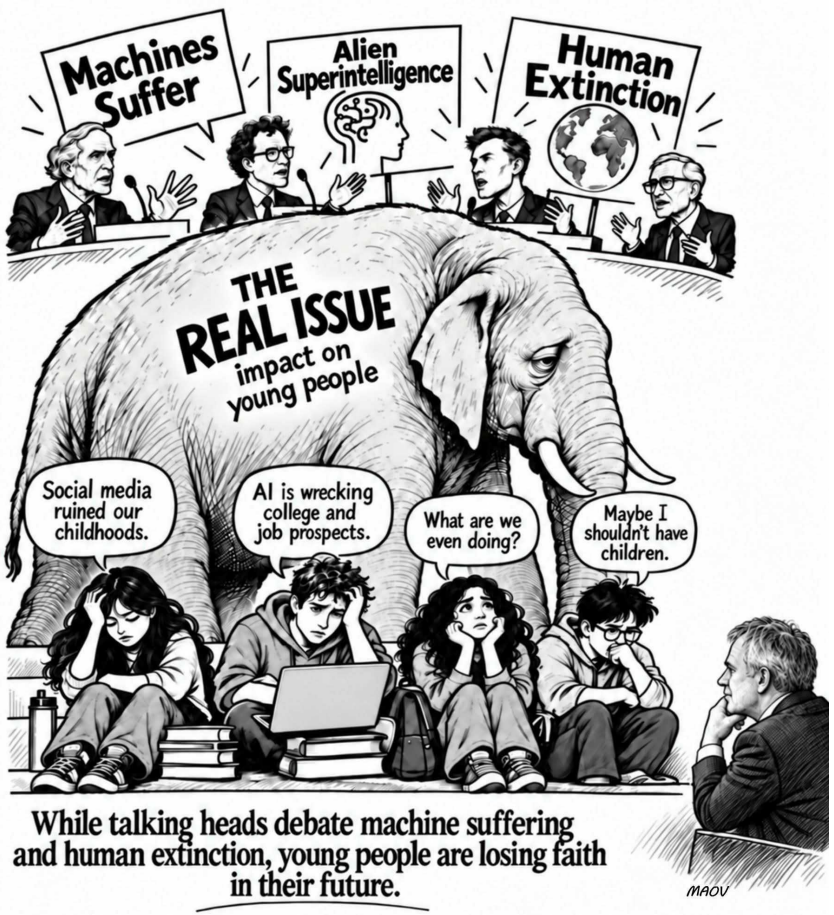
[Research ecosystem is overloaded, while 70-80% papers are non-reproducible]



Conference on Neural Information Processing Systems (NeurIPS); International Conference on Learning Representations (ICLR)

(GRAPHIC) V. PENNEY/*SCIENCE*; (DATA) ASSOCIATION FOR COMPUTATIONAL LINGUISTICS ROLLING REVIEW; CONFERENCE ON NEURAL INFORMATION PROCESSING SYSTEM; INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS; OPENACCEPT.ORG.

Oct 2026


Machines Suffer
Alien Superintelligence
Human Extinction
THE REAL ISSUE
impact on young people
Social media ruined our childhoods.
AI is wrecking college and job prospects.
What are we even doing?
Maybe I shouldn't have children.
While talking heads debate machine suffering and human extinction, young people are losing faith in their future.
MAOV

Main Section

Problem Solving Process

Receiving the problem ***{from client}***

- Client creates **requirements** document

- **Understanding the problem and scoping**
  - Developer creates **specification**

- **Solving**
  - Understand and implementing with tools for solving the problem: **Design docs, test plans**
  - Create solutions: **Code, Release**
  - Evaluate solutions: **Test logs**

- **Delivering** ***{to customers}***

A generic process involves:

- Customer:
  - requirements
- Developer:
  - specification
  - design
  - development (coding)
  - testing
  - release
- Customer:
  - evaluation
- acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Validating the Problem

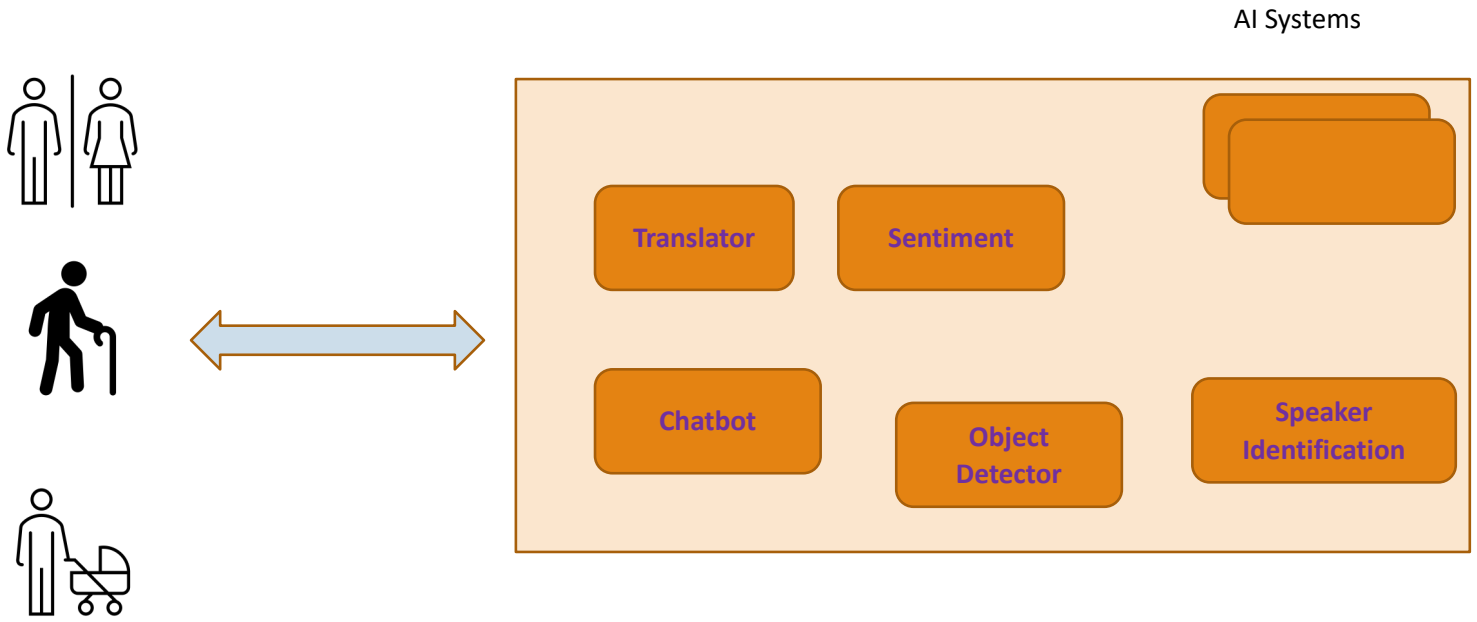
- **Objective**: Understanding the problem and scoping
- Think and articulate – what will be
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software, solution, documentation
  - Criteria for problem being solved: e.g., metrics

  - Considerations
    - Current practices/ processes
    - Why would someone use your solution to the problem?
    - **If you can't identify inputs, outputs, and evaluation metrics, the problem is most likely ill-defined computationally. Needs more work!**

Selecting Problems that Matter

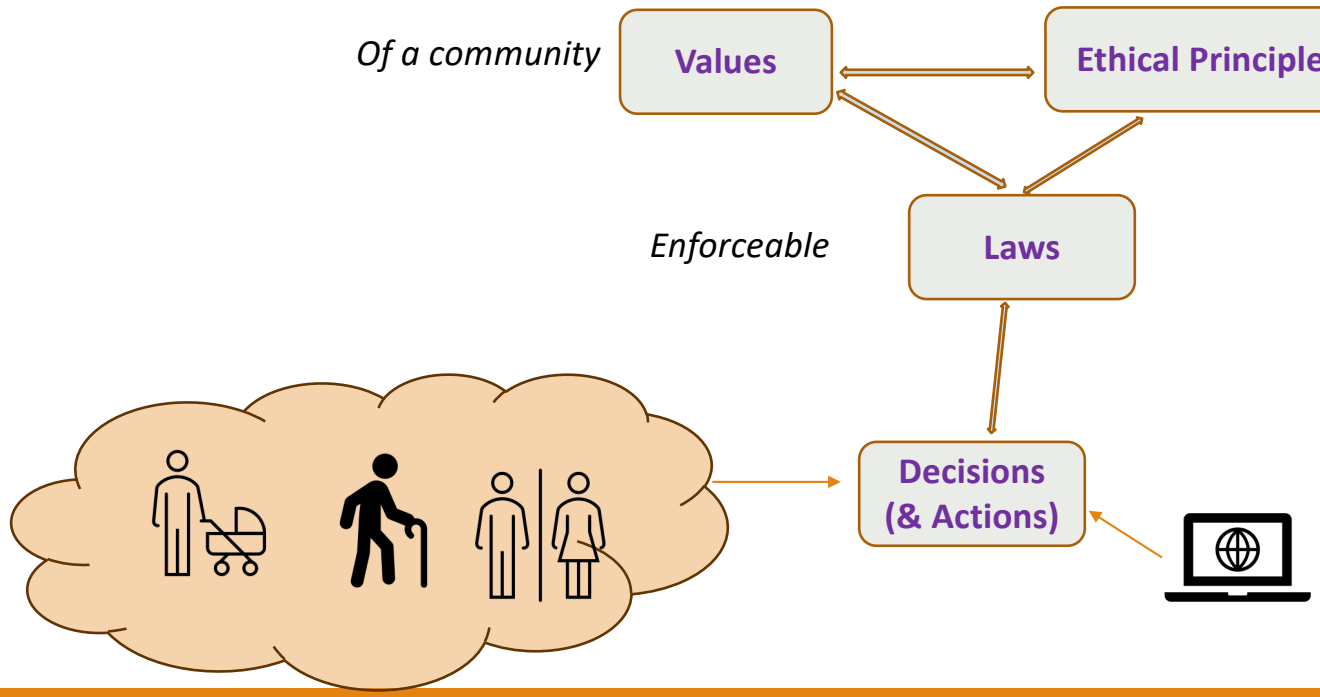
- Answer
  - For whom is the problem important? **[A lot of people]**
  - What happens if you solve it well? **[With change the status quo for the better]**
  - Do you have the inputs to get started? **[Yes? Otherwise, re-think problem or inputs]**
  - Do you have the resources/ are you best qualified to solve the problem? Do you need to augment the team? **[Yes? Otherwise, pass or get help]**

Technology and People



Trust: *Can people trust* ***new system*** *to perform capably, consistently, and with human values?*

Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can, humans or machines (AI) or in collaboration together, take decisions and actions following the same **values**, **ethical** principles and **laws ?***

Assumption: *they should follow same values, principles and laws.*

Concept – Verification, Validation



A generic process involves:

- Customer:
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- Developer:
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  - design
  - development (coding)
  - testing
  - release
- Customer:
  - evaluation
  - acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Concept – Functional v/s Non-functional

- Requirements – what the customer says they want
  - **Functional**: must have – typically, capabilities
  - **Non-functional**: good to have but may or may not be specified – typically, throughput, reliability,
- Specifications – what the developers say they implemented
  - **Functional**: must have – typically, capabilities
  - **Non-functional**: good to have but may or may not be specified – typically, throughput, reliability,

A generic process involves:

- Customer:
  - requirements
- Developer:
  - specification
  - design
  - development (coding)
  - testing
  - release
- Customer:
  - evaluation
  - acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Concept – Testing Types

- Unit testing – single developers, testing capability implemented
- System and integration testing – by a group of developers, testing system/ sub-system capabilities
- Acceptance testing – overall system's acceptance, **by the customer**

A generic process involves:

- Customer:
  - requirements
- Developer:
  - specification
  - design
  - development (coding)
  - testing
  - release
- Customer:
  - evaluation
  - acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Testing (AI) Chatbots – As Software

- Requirements – what the customer says they want
  - **Functional**: must have – typically, capabilities
  - **Non-functional**: good to have but may or may not be specified – typically, throughput, reliability,
- Specifications – what the developers say they implemented
- Testing types
  - Unit testing – single developers, testing capability implemented
  - System and integration testing – by a group of developers, testing system/ sub-system capabilities
  - Acceptance testing – overall system's acceptance, **by the customer**

Types of (AI) Software Evaluation

- **Technical evaluation**: is the (new) method accurate?
  - Concern is bug-free implementation
- **Business evaluation**: is the (new) method beneficial ?
  - Usually done in a small time horizon
  - Other factors may correlate with success or failure
- **Causal evaluation**: did the (new) method really work?
  - No other factor but this method contributed to business success

Exercise – 1 of 2

- Calculate 2 + 3 * 4 – 5 / 6 and express steps in English
  - Explain how you did it
- Calculate 2 + 3 * 4 – 5 / 6 with a calculator
  - Explain how the calculator did it

Exercise – 2 of 2

- Calculate 2 + 3 * 4 – 5 / 6 and express steps in English
- Explain which LLMs do correctly and how did you infer that

Llama-3-70b-Groq, vis Poe

I'd be happy to help with the calculation.

To evaluate the expression 2 + 3 * 4 – 5 / 6, we need to follow the order of operations (PEMDAS):

Multiply 3 and 4: 3 * 4 = 12
Divide 5 by 6: 5 / 6 = 0.83 (approximately)
Add 2 and 12: 2 + 12 = 14
Subtract 0.83 from 14: 14 - 0.83 = 13.17

Therefore, the final answer is approximately 13.17.

Bot image for Deepseek-V3-FW

Deepseek-V3-FW
Poe

Certainly! Let's solve the expression step by step following the order of operations (PEMDAS/BODMAS: Parentheses/Brackets, Exponents/Orders, Multiplication and Division (left to right), Addition and Subtraction (left to right)).
Expression:

2 + 3 * 4 – 5 / 6

Multiplication (3 * 4):
Multiply 3 and 4 → 12.
Now: 2 + 12 − 5 / 6.

Division (5 / 6):
Divide 5 by 6 → ~0.833 (repeating).
Now: 2 + 12 − 0.833.

Addition (2 + 12):
Add 2 and 12 → 14.
Now: 14 − 0.833.

Subtraction (14 − 0.833):
Subtract 0.833 from 14 → ~13.167.

Final Answer:

≈ 13.167

(Note: Exact value is 14−56=846−56=796≈13.16714−65=684−65=679≈13.167).

Steps are summarized:

Multiply 3 × 4 = 12.
Divide 5 ÷ 6 ≈ 0.833.
Add 2 + 12 = 14.
Subtract 14 − 0.833 ≈ 13.167.

Problem Example 1: Student Meals Reco

- **Problem**: build a system to help students choose meals during study days
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - Recommendation methods
- Trustworthiness ecosystem
  - Values, ethical principles, laws
  - Users impacted

Problem statement criteria

- Are focused on the user (avoid using “we” or “the product”)
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Problem Example 2: Nobel Prizes

- **Problem**: build a system to contextualize about Nobel Prizes wins
- Validating the problem:
  - The inputs: e.g. data, current process
  - The outputs (deliverables): e.g., software (app)
  - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
  - **Identifying topics and trends**
- Trustworthiness ecosystem
  - Values, ethical principles, laws -
  - Users impacted – ***students making future decisions based on trends ?***

Problem statement criteria

- Are focused on the user (avoid using "we" or "the product")
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Nobel Prizes: Non-functional Requirements/ Specification

- Winners background
  - Name, gender, age, country, institution
  - Should or shouldn't be given ? Any disclaimer
- What do the awards in a year tell about
  - The past ?
  - The future ?
- Themes
  - Terminology changes – how to handle them

Snapshot from Different Sections

Section 1	Section 2	Section 4
1. students on scooters (harm) 2. Students managing money in college 3. Students finding housing (off-campus) 4. Students managing assignments **5. (all) Sidewalks uneven** 6. Students managing finances – 1st year 7. Students: saving meal swipes (7!)	**1. Students: textbook overprices and requirements** 2. (All) drunk driving at 5-points **3. Students: on-campus laundry** 4. Students: USC bus trackers are unreliable 5. Students: non-reserved parking 6. Students: double-deck parking	1. Students: pedestrian safety, accessibility 2. Students: campus parking and transportation 3. {} – problem not noted 4. Students: college tuition payment **5. Students: time management and resources**

Assignment 8: Same for All in a Team

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see ***problems that matter***
- Write a problem statement
  - The group will have one testing plan
  - Add your statement to your GitHub in a sub-folder called 'assignment-8'
    - Do mention all the students in your group

- Work with your group
  - If you have a group, sit with them
  - If you don't, join an existing group.
- Discuss a system setup
  - Inputs and outputs
  - Metrics
- Discuss **evaluation**
  - Functional specification
  - Non-functional specification
- Write test plans
  - A test cases has a sample inputs and expected outputs
  - Q: How many testcases to create ?

See sample testcases at: https://github.com/biplav-s/course-cmw-f26/tree/main/software-dev

Concluding Section

Conclusion

We talked about

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	**All extensions end**

Ask Me Anything